Celina(Yue) Jiang

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EDUCATION

University of Pennsylvania **Philadelphia, PA**
Master of Engineering in Chemical and Biomolecular Engineering 1/2026 – Expected 5/2027

- **Coursework:** Chemical Kinetics and Reactor Design, Thermodynamics, Transport Process

Technion, Israel Institute of Technology (English-medium education) **Shantou, China**
Bachelor of Science in Chemical Engineering 8/2021 - 7/2025

- **Coursework:** Separation Operations, Process Dynamics & Control, Plant Design, Process Design

SKILLS

Technical Skills: Python, MATLAB, Origin, ChemDraw, UNISIM, GaussView, SAXS, FTIR, POM, ICP-OES, GC, FT-ICR MS, STATA
Languages: English (Proficient), Chinese (Native)

RESEARCH EXPERIENCE

Fabricating Nanofiltration Membranes Using Self-assembled Materials for Critical Mineral Recovery Application
Graduate Research Assistant, Supervisor: Prof. Chinedum Osuji 8/2026 – Present

- Developed self-assembled nanofiltration membranes for direct lithium extraction (DLE) and critical mineral recovery
- Performed SAXS, FTIR, and polarized optical microscopy to characterize membrane structure and evaluate the impact of cross-linker composition on material properties
- Investigated Li^+/Na^+ and Li^+/K^+ transport and sorption using dialysis diffusion and ICP-OES to evaluate ion selectivity
- Conducted water uptake and tensile testing to assess membrane stability and mechanical performance

Formation and Processing of Carbon Hollow Fiber Membranes (CHFMs) for Gas Separation
Undergraduate Research Assistant, Supervisor: Prof. Xuezhong He 4/2024 – 7/2025

- Fabricated high-performance CHFMs from cellulose via optimized spinning, drying, and carbonization conditions to enhance the separation efficiency of carbon molecular sieve membranes for CO_2 capture
- Developed and tested hollow fiber membranes, analyzing separation properties for gases including CO_2 using membrane separation device
- Using optical microscopy and gas chromatograph, identified four critical causes for membrane rupture: bubble formation, unstable dope, unstable bore, incomplete solidification

Exploring the Influence of Sludge Organic Components on Dewaterability
Undergraduate Research Assistant, Supervisor: Prof. Keke Xiao 7/2023 – 6/2024

- Investigated how organic components affect sludge dewaterability, analyzing water retention and release using capillary suction time and specific resistant filtration
- Analyzed sludge organic matter using Fourier transform ion cyclotron resonance mass spectrometry (FT-ICR MS) and correlated molecular characteristics with dewatering behavior

PUBLICATIONS

X. Liu, **Y. Jiang**, et.al (2026). Superation and Purification Technology (140617)

R. Wang, M. Nabi, **Y. Jiang**, K, Xiao (2024). *ELSEVIER Environmental Research,* 262 (119900)

M. Nabi, Y. Jiang, W. Ding, R. Wang, **Y. Jiang**, et.al (2025). *Chemical Engineering Journal,* 506 (160187)

Y. Lin, Z. Xu, Y. Jiang, **Y. Jiang**, K. Xiao (2025). *Water* 2025*, 17, 2019*

WORK EXPERIENCE

Student Representative | UPenn Chemical and Biomolecular Engineering 10/2026 – present

- Advocated for international students by proposing recommendations regarding Optional Practical Training (OPT) and Curricular Practical Training (CPT).

Student Advisory Group for the Environment (SAGE) - Member 9/2026 - present

- Proposed environmental sustainability recommendations to support waste reduction and recycling initiatives across campus.
- Participated in the Club Composting initiative to explore composting solutions for student events, including stakeholder engagement, feasibility assessment, and potential pilot program development.

Guangdong-Technion Technology and Entrepreneurship Collaboration 5/2023 – 7/2025

- Organized industry expert events and facilitated industry–academia collaboration, connecting companies with university research projects to promote engineering innovation.
- Assisted with patent applications and managed laboratory equipment inventory, supporting research commercialization and efficient lab operations.

Student Representative | Technion, Israel Institute of Technology 10/2022 – 7/2025

- Collected and consolidated student feedback on academic matters, communicating concerns and recommendations to undergraduate administration.
- Served as a liaison between students, faculty, and university administrators, facilitating discussions with professors and supporting the resolution of student concerns.

Career and Learning Center, GTIIT – Peer Tutor 8/2022 – 7/2023

- Led weekly tutoring sessions for 20 first-year students in Physics, Mathematics, and Chemistry